

JavaScript Switch Case Statement – Complete Reference Guide

1. Switch Expression

Switch expression can evaluate to Number, String, Boolean, null, undefined, Object Reference, Symbol, BigInt and a single-character string (JavaScript has no separate char type).

```
let day = 2;
switch(day){
  case 1: console.log("Mon"); break;
  case 2: console.log("Tue"); break;
}
```

2. Case Value Type

Case values should match the switch expression type because JavaScript uses strict equality (===).

```
let value = "10";
switch(value){
  case 10: console.log("Number"); break;
  case "10": console.log("String"); break;
} // Output: String
```

3. break Statement

break is optional but recommended. Without it, execution falls through.

```
switch(1){
  case 1:
    console.log("One");
  case 2:
    console.log("Two");
}
// Output:
// One
// Two
```

4. default Statement

default executes when no case matches.

```
switch("Blue"){
  case "Red":
    console.log("Stop");
    break;
  default:
    console.log("Unknown");
}
```

5. Comparison Rule

switch always compares using strict equality (===). Value and type must both match.

```
10 === "10"    // false
true === 1     // false
```

6. Execution Flow

Evaluate expression → Compare cases top to bottom → Execute first matching case → Exit on break → Otherwise continue (fall-through) → Execute default if nothing matches.

7. Multiple Cases

Multiple case labels can share one block.

```
switch(day){
  case 6:
  case 7:
    console.log("Weekend");
    break;
}
```

8. Expressions

Both switch expression and case labels may be expressions.

```
let a = 10;
switch(a + 5){
  case 15:
    console.log("Matched");
    break;
}
```

9. Best Use Cases

Use switch for exact matching such as menus, user roles, API status codes, days, months and calculator operators.

10. Avoid for Ranges

Use if...else instead of switch for ranges.

```
// Recommended
if(marks >= 90){
  console.log("A");
}else if(marks >= 80){
  console.log("B");
}
```

Interview Tips

- switch uses strict equality (===).
- break exits only the switch block.
- Missing break causes fall-through.
- default is optional but recommended.
- Best for exact value matching, not ranges.
- JavaScript has no separate char data type.